# Introduction to Biophysics II: Quantum physics & biology

## Lecture 06: Energy transfer in photosynthesis - I

Energy transfer: examples in photosynthetic proteins
Förster Theory:
- Examples of photosynthetic proteins
- Single molecule:
  - Nuclear wave function, vibrations, Franck-Condon, Debye-Waller
  - Absorption and emission: Stokes shift
- Dimer: noninteracting and interacting
- Förster Resonance Energy Transfer: solving Time-Dependent Schrödinger
- Density of states, spectral overlap

Reading:
Modern Optical Spectroscopy, chapter 4, 7
Additional:
Quantum Effects in Biology, chapter 3

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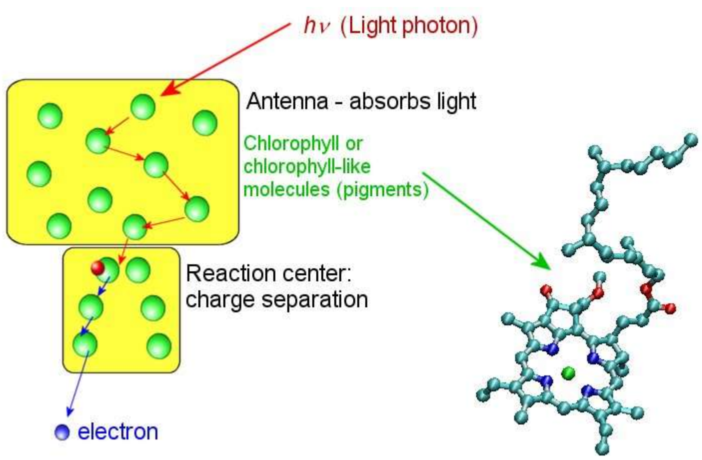
# Antenna and reaction center

- A large variety of different photosynthetic systems (plants, bacteria, algae) share two common elements: light-harvesting **antenna** and **reaction center** (RC).

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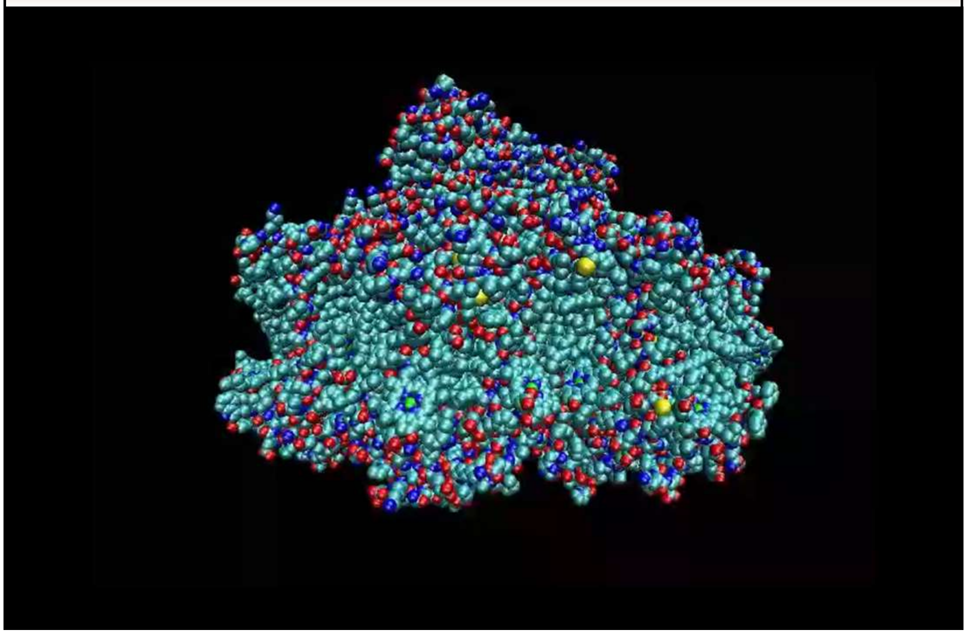
# Photosystem I: example



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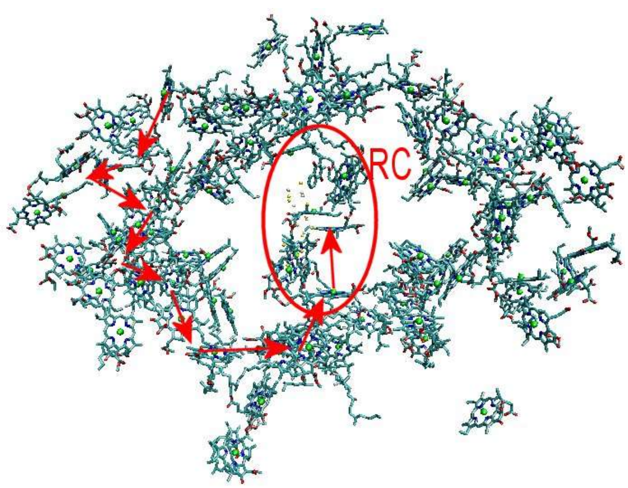
# Antenna and reaction center

- Example: photosystem I (PS I)




96 Chl *a* pigments capture light and transfer excitation energy to the reaction center
Quantum efficiency ~100% !

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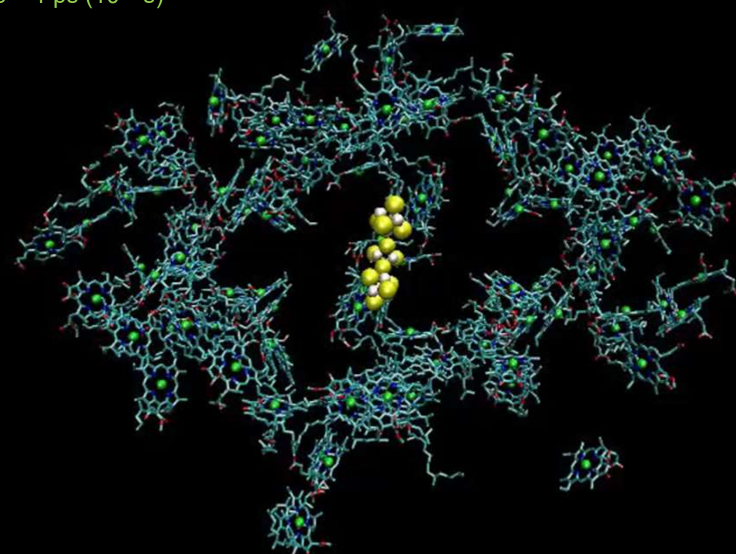
## PS I under 1,000,000,000,000X time microscope

1s = 1 ps ($10^{-12}$s)



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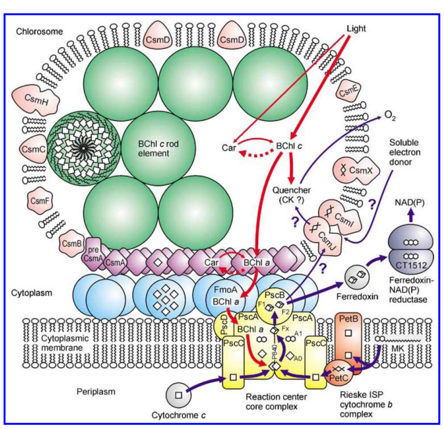
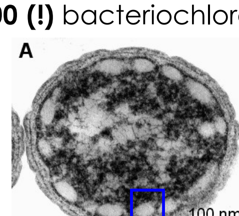
## Antenna and reaction center

- Example: chlorosomes

Common light-harvesting antenna complex in green bacteria

- The most efficient antenna known: **200,000 (!)** bacteriochlorophylls




*Chlorobium tepidum*

*Hot spring, Yellowstone*

From Frigaard, N. *et al.* 2003 *Photosyn. Res.* **78:** 93-117

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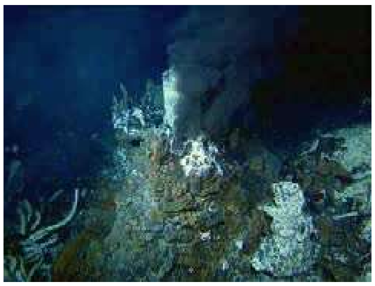
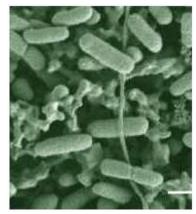
# Antenna and reaction center

- Example: chlorosomes

The champions of low-light photosynthesis:
need only 1 photon absorbed by a *BChl* in 8 hours to survive

Hydrothermal vent
("black smoker")





Depth: 2.4 km; Vent: 350ºC; Water: 2°C

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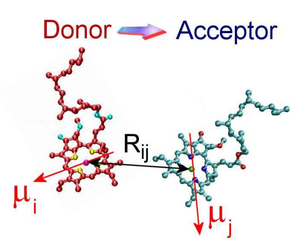
# Excitation energy transfer

The excitation energy can be transferred from one molecule to another:

1. The *donor* molecule (*D*) is promoted into an excited state due to light absorption or energy transfer
2. Due to coupling with a nearby *acceptor (A)* molecule, the donor molecule can transition into the ground state and *simultaneously,* the acceptor molecule is promoted into an *excited* state:




$$D^* + A = D + A^*$$

Where $^*$ denotes an excited state

- Energy transfer efficiency depends on the *coupling* between the molecular transitions of *D* and *A*.
- Energy must be conserved!

*Resonance energy transfer – the* absorption spectrum of *A* overlaps with the emission spectrum of *D*.
*Note: there is* *no emission* or *absorption* in this process; energy is directly transferred from D to A

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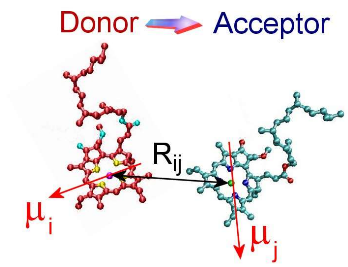
# Energy transfer between molecules

Consider 2 molecules with transitions $\mu_{i,j}$




Energy transfer: quantum of energy is exchanged → the molecules must have the same energies.

For long-living molecules, transition energies are well-defined, and overlap is very improbable.

→ **Need to spread their energies through molecular and bath vibrations**

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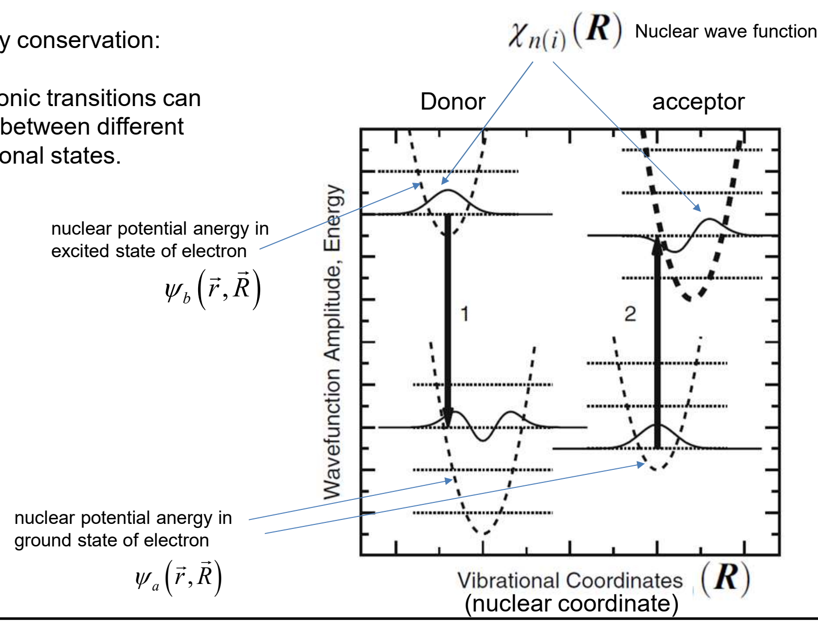
# Energy transfer: energy conservation

Energy conservation:

Electronic transitions can occur between different vibrational states.

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# Nuclear vibrations/rotations

***Born-Oppenheimer approximation***: electrons are light, nuclei are heavy, can assume that electrons adapt "instantaneously" to a new nuclear configuration

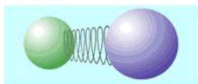
$$\Psi(\boldsymbol{r},\boldsymbol{R}) \approx \psi_i(\boldsymbol{r},\boldsymbol{R})\chi_{n(i)}(\boldsymbol{R})$$

Nuclear coordinates

Nuclear wavefunction

electronic wave function

Consider nuclear part: $\chi(\mathbf{R})$.

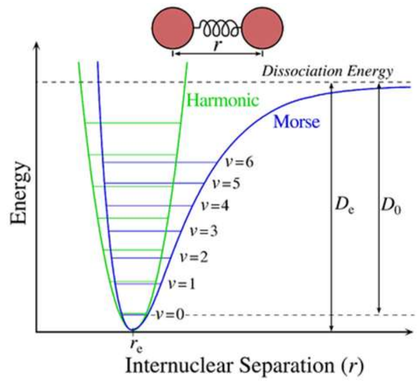


In the case of a diatomic molecule, **R** is simply the distance between atoms.

Vibration energies are quantized!

When **R** is not far from equilibrium, it can be approximated as a parabola, i.e., a harmonic potential

Note: The electronic part of $\psi$ depends *parametrically* on nuclear coordinate

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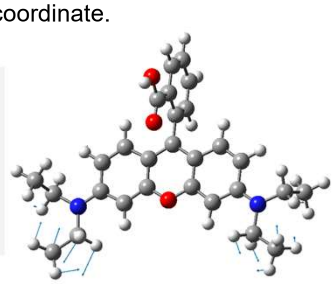
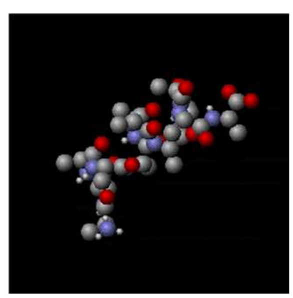
# Nuclear vibrations/rotations

For 2+ atom molecules vibrations are more complex

We can analyze them as a superposition of normal modes: orthogonal vibrations whose superposition can represent any complex vibration. The potential for each could be still represented as a harmonic function of *generalized* nuclear coordinate.





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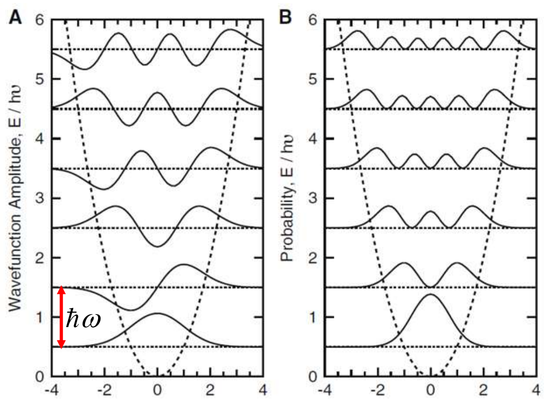
# Nuclear vibrations/rotations

Consider the nuclear part: $\chi(\mathbf{R})$, harmonic oscillator.

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}k\hat{x}^2 = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \quad \rightarrow \quad \omega = \sqrt{\frac{k}{m}}$$




Note: any potential near minimum approaches harmonic potential

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# Nuclear vibrations/rotations

Consider wave function: $\Psi(\boldsymbol{r}, \boldsymbol{R}) \approx \psi_i(\boldsymbol{r}, \boldsymbol{R})\chi_{n(i)}(\boldsymbol{R})$

dipole operator: $\tilde{\mu} = \tilde{\mu}_{el} + \tilde{\mu}_{nuc} = \sum_i e\boldsymbol{r}_i + \sum_j z\boldsymbol{R}_j$

Nuclear charge

Transition dipole for 1 electron (transition from state *a* to state *b*):

$$\vec{\mu}_{ba,mn} = \left\langle \psi_b\left(\vec{r},\vec{R}\right)\chi_m\left(\vec{R}\right)\middle|\tilde{\mu}_{el}\middle|\psi_a\left(\vec{r},\vec{R}\right)\chi_n\left(\vec{R}\right)\right\rangle + \left\langle \psi_b\left(\vec{r},\vec{R}\right)\chi_m\left(\vec{R}\right)\middle|\tilde{\mu}_{nuc}\middle|\psi_a\left(\vec{r},\vec{R}\right)\chi_n\left(\vec{R}\right)\right\rangle$$

$$= e\int \chi_m^*\left(\vec{R}\right)\chi_n\left(\vec{R}\right)\left[\int \psi_b^*\left(\vec{r},\vec{R}\right)\vec{r}\psi_a\left(\vec{r},\vec{R}\right)d\vec{r}\right]d\vec{R}$$

$$+ z\int \chi_m^*\left(\vec{R}\right)\vec{R}\chi_n\left(\vec{R}\right)\underbrace{\boxed{\left[\int \psi_b^*\left(\vec{r},\vec{R}\right)\psi_a\left(\vec{r},\vec{R}\right)d\vec{r}\right]}}_{=?}d\vec{R}$$

$$\vec{\mu}_{ba,mn} = e\int \chi_m^*\left(\vec{R}\right)\chi_n\left(\vec{R}\right)\left[\int \psi_b^*\left(\vec{r},\vec{R}\right)\vec{r}\psi_a\left(\vec{r},\vec{R}\right)d\vec{r}\right]d\vec{R}$$

Problem: electronic wavefunction depends on **R**, cannot separate 2 integrals

Workaround: average $\langle \psi_b(\boldsymbol{r}, \boldsymbol{R})|\tilde{\mu}_{el}|\psi_a(\boldsymbol{r}, \boldsymbol{R})\rangle$ over a range of **R** in the initial state, denote $\boldsymbol{U}_{ba}$:

$\boldsymbol{\mu}_{ba,mn} \approx \langle \chi_m(\boldsymbol{R})|\chi_n(\boldsymbol{R})\rangle \overline{\boldsymbol{U}}_{ba}$ ← "**Condon approximation**"

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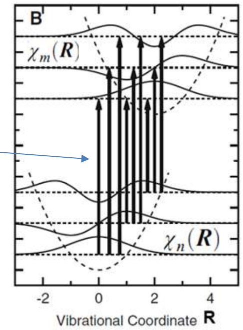
# Nuclear vibrations/rotations

**Condon approximation** $\mu_{ba,mn} \approx \langle \chi_m(\mathbf{R}) | \chi_n(\mathbf{R}) \rangle \overline{U}_{ba}$

Final ($\psi_b$) state

Initial ($\psi_a$) state

electron is lighter than a nucleus →
During the transition, nuclei do not shift.
Transitions are *vertical*




$\overline{\mathbf{U}}_{ba}$ is average of our usual transition dipole $\mu_{ba}$ over nuclear coordinates of the initial state

$$\overline{U}_{ba} = \overline{\langle \psi_b(\mathbf{r},\mathbf{R}) | \tilde{\mu}_{el} | \psi_a(\mathbf{r},\mathbf{R}) \rangle}$$

$\langle \chi_m | \chi_n \rangle$ - *nuclear wavefunctions overlap integral*

$|\mu_{ba,mn}|^2$ - Contribution of a particular vibronic transition to the dipole strength

$|\langle \chi_m | \chi_n \rangle|^2$ - **Franck-Condon factor.**

*Note:* <$\chi_m$| (left) is for the *excited state,* |$\chi_n$> (right) *is for the ground state!*

**Franck-Condon** principle:
during an electronic transition, a change from one vibrational energy level to another will be more likely to happen if the two vibrational wave functions overlap more significantly.

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# Nuclear vibrations/rotations

Condon approximation $\mu_{ba,mn} \approx \langle \chi_m(\mathbf{R}) | \chi_n(\mathbf{R}) \rangle \overline{U}_{ba}$

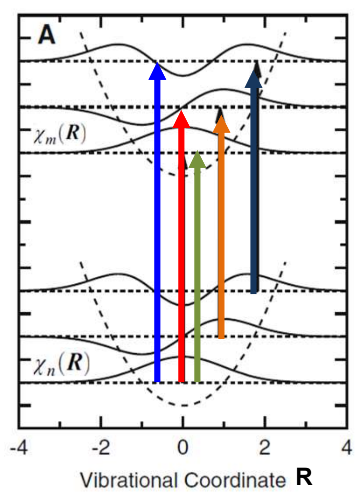
$|\langle \chi_m | \chi_n \rangle|^2$ - **Franck-Condon factor**

<$\chi_m$| (left) is for the *excited state*
|$\chi_n$> (right) *is for the ground state!*

**Example:**
equilibrium nuclear configuration does not change upon electronic transition.

Which transitions are most intense?

What will the absorption spectrum look like?

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# Nuclear vibrations/rotations

Condon approximation $\mu_{ba,mn} \approx \langle \chi_m(\mathbf{R}) | \chi_n(\mathbf{R}) \rangle \overline{U}_{ba}$

$|\langle \chi_m | \chi_n \rangle|^2$ - **Franck-Condon factor**

**Example:**
equilibrium nuclear configuration changes upon electronic transition

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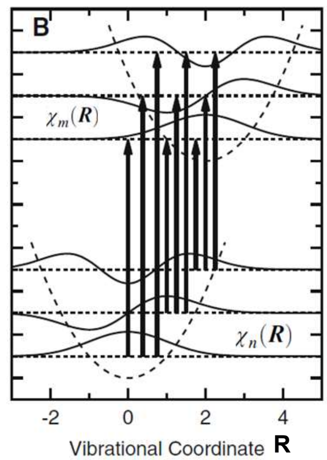


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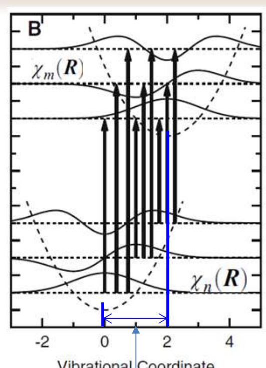
# Nuclear vibrations/rotations

Condon approximation $\mu_{ba,mn} \approx \langle \chi_m(\mathbf{R}) | \chi_n(\mathbf{R}) \rangle \overline{U}_{ba}$

$|\langle \chi_m | \chi_n \rangle|^2$ - **Franck-Condon factor**

Approximate Franck-Condon factors can be obtained by using the wavefunctions of harmonic oscillator assuming that displacement of equilibrium coordinate does not affect the "spring constant", i.e. ħω is the same for ground and excited states.




Define *dimensionless displacement* (diatomic):

$$\Delta = 2\pi\sqrt{m_r \nu / h}\,\boxed{(b_e - b_g)}$$

Change in bond length

ν - frequency of vibration

Reduced mass of nuclei, $1/m_r = 1/m_1 + 1/m_2$

Define *coupling strength* ***(Huang-Rhys factor)*** $S = \frac{1}{2}\Delta^2$

Can derive: $\boxed{|\langle \chi_m | \chi_0 \rangle|^2 = \frac{S^m \exp(-S)}{m!}}$ **= Franck-Condon factors**
Transition from $\psi_a \chi_0$ to $\psi_b \chi_m$

Can derive a similar expression for transitions from other than 0 level

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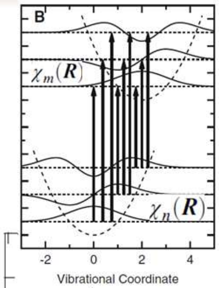
# Nuclear vibrations/rotations

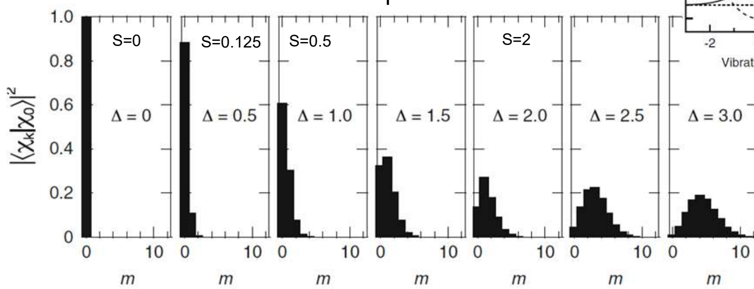
$$\boldsymbol{\mu}_{ba,mn} \approx \langle \chi_m(\boldsymbol{R}) | \chi_n(\boldsymbol{R}) \rangle \overline{\boldsymbol{U}}_{ba}$$

$$\boxed{|\langle \chi_m | \chi_0 \rangle|^2 = \frac{S^m \exp(-S)}{m!}} \qquad S = \frac{1}{2}\Delta^2$$

$$\Delta = 2\pi\sqrt{m_r \upsilon / h}\,(b_e - b_g)$$




Example: transition strength from the lowest to higher vibrational levels as a function of displacement:




Note: A molecule with $N$ atoms has 3N-6 vibrational modes (3N-5 for a linear molecule). Can write nuclear wavefunction as a product of wavefunctions of individual modes (similarity with Fourier series)

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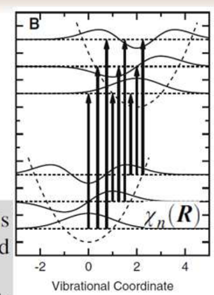
# Franck-Condon factors

Just for reference: one can calculate overlaps between arbitrary vibrational levels as shown below (from Box 4.13 of the textbook):




Note that the vibrational wavefunctions in the bra and ket portions of this expression implicitly pertain to different electronic states. We have dropped the indices *a* and *b* for the electronic states to simplify the notation.

Overlap integrals for other combinations of vibrational levels can be built from $\langle \chi_0 | \chi_0 \rangle$ by the recursion formulas:

$$\langle \chi_{m+1} | \chi_n \rangle = (m+1)^{-1/2} \left\{ n^{1/2} \langle \chi_m | \chi_{n-1} \rangle - S^{1/2} \langle \chi_m | \chi_n \rangle \right\} \qquad \text{(B4.13.2}a\text{)}$$

and

$$\langle \chi_m | \chi_{n+1} \rangle = (n+1)^{-1/2} \left\{ m^{1/2} \langle \chi_{m-1} | \chi_n \rangle + S^{1/2} \langle \chi_m | \chi_n \rangle \right\}, \qquad \text{(B4.13.2}b\text{)}$$

with $\langle \chi_m | \chi_{-1} \rangle = \langle \chi_{-1} | \chi_m \rangle = 0$. For the overlap of the lowest vibrational level

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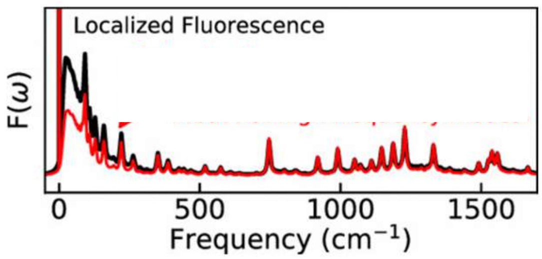
## Linewidth broadening

Lecture 04

**Coupling with "bath":** molecule is embedded into protein which has its own vibrational modes, acoustic modes are low-frequency almost continuous in spectrum




Chl *a* fluorescence
58 vibrational modes
+ bath

Mike Reppert: . *Phys. Chem. B* 2020, 124, 45, 10024–10033

Vibrations lead to ***homogeneous*** broadening, i.e. they show even when there is only one light-absorbing molecule excited.

https://pubs.acs.org/doi/abs/10.1021/acs.jpcb.0c05789

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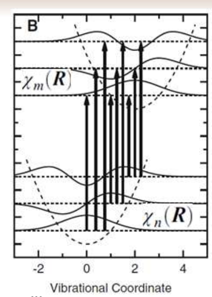
## Nuclear vibrations/rotations

$$\boldsymbol{\mu}_{ba,mn} \approx \langle \chi_m(\boldsymbol{R}) | \chi_n(\boldsymbol{R}) \rangle \overline{\boldsymbol{U}}_{ba}$$

$$\boxed{|\langle \chi_m | \chi_0 \rangle|^2 = \frac{S^m \exp(-S)}{m!}}$$




$<\chi_m|\chi_0>$ gives you a "projection" of $\chi_0$ of ground state onto each wavefunction of excited state $\chi_m$, i.e.

$$\chi_o = \sum_m \langle \chi_m | \chi_o \rangle \chi_m$$

Multiply and integrate: $\times \left| \chi_o \right\rangle$

$$\langle \chi_o | \chi_o \rangle = \sum_m \langle \chi_m | \chi_o \rangle \langle \chi_m | \chi_o \rangle$$

$$\sum_m \left| \langle \chi_m | \chi_o \rangle \right|^2 = 1$$

Solutions for harmonic oscillator wave functions constitute complete orthonormal set: can express any other function as a superposition (like Fourier series)
$\langle \chi_m | \chi_o \rangle$ ← Like 'Fourier trick'

I.e. sum of all Franck-Condon factors =1,

$\left| \langle \chi_o | \chi_o \rangle \right|^2$ = The ratio of 0-0 transition strength to the total transition dipole strength

$\left| \langle \chi_o | \chi_o \rangle \right|^2 = \exp(-S)$ ← **Debye-Waller factor** (error in book p.187, exp(~~-S/2~~))

*Note: this is an overlap between vibrational ground states of different electronic states!*

*Note: assumed that vibration frequencies in excited and ground states are the same*

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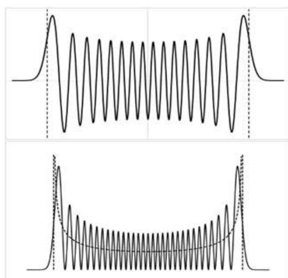
# The Stokes Shift

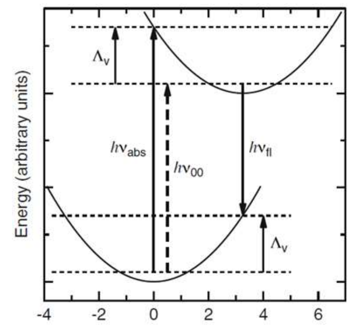
Absorption and emission spectra maximize at different wavelengths (transition energy).

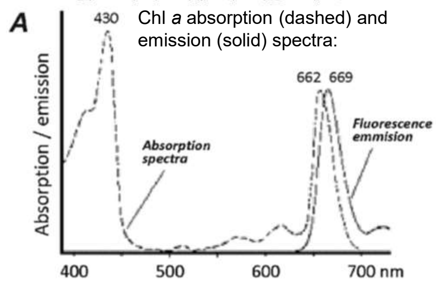
The shift is called **Stokes shift**

Note: the higher the quantum state, the higher the probability of finding the particle at the classical 'turning points" (on the potential energy curve)



Wavefunction (top) and probability density (bottom) for the $n = 30$ excited state of the quantum harmonic oscillator.

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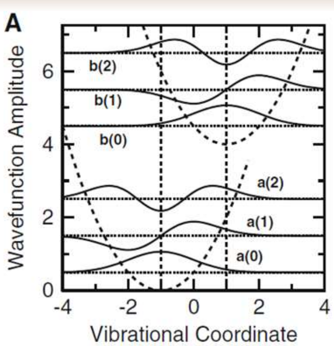
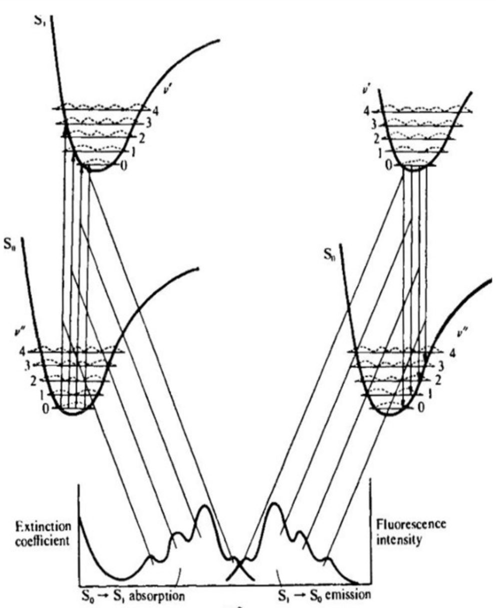
# Mirror image rule




Transition down in our simple model involves the same displacement; thus, Franck-Condon factors for fluorescence are the same as for absorption.

→ **absorption and emission spectra are (*almost*) mirror images of each other.**

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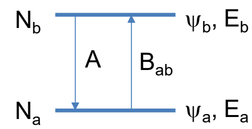
# Mirror image rule

Note: absorption and emission spectra are **NOT *exact*** mirror images.

Recall:

$$\frac{dN_b}{dt} = -N_b A - N_b B_{ba}\rho(\omega_0) + N_a B_{ab}\rho(\omega_0)$$




Emission $A = \dfrac{\omega_0^3 |\boldsymbol{p}|^2}{3\pi\epsilon_0\hbar c^3}$.

Absorption $B_{ab} = B_{ba} = \dfrac{\pi}{3\epsilon_0\hbar^2}|\boldsymbol{p}|^2$.

Emission has an additional factor $\omega_0^3$ and its shape also depends on the units! This will modify the intensity distribution of vibronic bands in the emission spectrum compared to absorption!

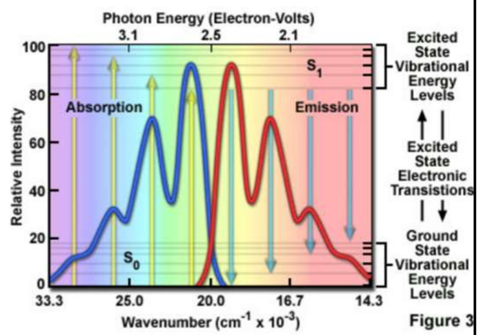



Figure 3

https://www.chem.uci.edu/~dmitryf/manuals/Fluorescence%20Excitation%20and%20Emission%20Fundamentals.pdf

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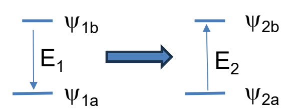
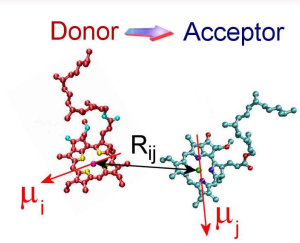
# Energy transfer between molecules.

Donor
Molecule 1

Acceptor
Molecule 2




***Förster Resonance Energy Transfer: FRET (1946)***

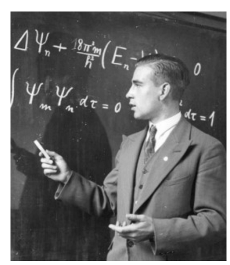
Approach: an excited state of complex (AB)* can be described as a super-molecule with fixed nuclear positions and two valence electrons moving in supramolecular potential.

Molecule 1 excited: $\Psi_1 = \Psi_{1b}\Psi_{2a}$

Molecule 2 excited: $\Psi_2 = \Psi_{1a}\Psi_{2b}$

The initial and final states are delocalized over both molecules:

$$\Psi(t) = c_1\Psi_1(t) + c_2\Psi_2(t)$$



Theodor Förster
1910 - 1974

*sometimes (biologists) call it fluorescence resonance energy transfer (FRET) – not historically nor scientifically correct, there is no fluorescence involved.*

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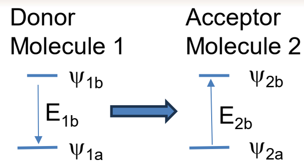
# Energy transfer between molecules.




$$\Psi_1 = \Psi_{1b}\Psi_{2a}$$
$$\Psi_2 = \Psi_{1a}\Psi_{2b}$$
$$\Psi(t) = c_1(t)\Psi_1(t) + c_2(t)\Psi_2(t)$$

Hamiltonian: $\tilde{\mathrm{H}} = \tilde{\mathrm{H}}_1 + \tilde{\mathrm{H}}_2 + \underbrace{\tilde{\mathrm{H}}_{21}}_{\textit{interaction}}$

Let the energy of the ground state $\Psi_0 = \Psi_{1a}\Psi_{2a}$ be zero. With no interaction:

$$E_1 = \left\langle \Psi_1 | \hat{H}_1 + \hat{H}_2 | \Psi_1 \right\rangle = \left\langle \Psi_{1b}\Psi_{2a} | \hat{H}_1 + \hat{H}_2 | \Psi_{1b}\Psi_{2a} \right\rangle$$
$$E_1 = \left\langle \Psi_{1b}\Psi_{2a} | \hat{H}_1 | \Psi_{1b}\Psi_{2a} \right\rangle + \left\langle \Psi_{1b}\Psi_{2a} | \hat{H}_2 | \Psi_{1b}\Psi_{2a} \right\rangle$$
$$E_1 = \left\langle \Psi_{1b} | \hat{H}_1 | \Psi_{1b} \right\rangle \left\langle \Psi_{2a} | \Psi_{2a} \right\rangle + \left\langle \Psi_{2a} | \hat{H}_2 | \Psi_{2a} \right\rangle \left\langle \Psi_{1b} | \Psi_{1b} \right\rangle = E_{1b}$$

Similarly: $E_2 = E_{2b}$

Need interaction $H_{21}$ for energy transfer ($c_1$ and $c_2$ will then depend on time)

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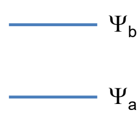
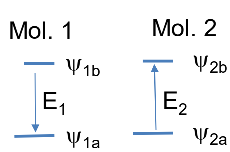
# Energy transfer & transition between states

Transition between states (Lecture 3)




$$\Psi(t) = c_a\Psi_a(t) + c_b\Psi_b(t)$$
$$H = H^0 + H'(t).$$
$$\hat{H}\Psi = i\hbar\frac{\partial\Psi}{\partial t}$$
$$c_b^{(1)} = -\frac{i}{\hbar}\int_0^t H'_{ba}(t')e^{i\omega_0 t'}\,dt'$$
$$\omega_0 \equiv \frac{E_b - E_a}{\hbar}$$

Energy transfer between states




1 excited: $\Psi_1 = \Psi_{1b}\Psi_{2a}$

2 excited: $\Psi_2 = \Psi_{1a}\Psi_{2b}$

$$\Psi(t) = c_1\Psi_1(t) + c_2\Psi_2(t)$$
$$\tilde{\mathrm{H}} = \tilde{\mathrm{H}}_1 + \tilde{\mathrm{H}}_2 + \tilde{\mathrm{H}}_{21}$$
$$\hat{H}\Psi = i\hbar\frac{\partial\Psi}{\partial t}$$

Math is exactly the same as in lecture 3!

$$C_2^{(1)} = -\frac{i}{\hbar}\int_0^t \tilde{H}_{21}\exp\left(i\frac{E_2 - E_1}{\hbar}t\right)dt$$

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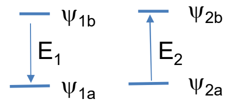
# Förster Energy Transfer

$$\Psi(t) = c_1 \Psi_1(t) + c_2 \Psi_2(t)$$

$$\omega_0 \equiv \frac{E_2 - E_1}{\hbar} \equiv \frac{E_{21}}{\hbar}$$




1 excited: $\Psi_1 = \Psi_{1b}\Psi_{2a}$

2 excited: $\Psi_2 = \Psi_{1a}\Psi_{2b}$

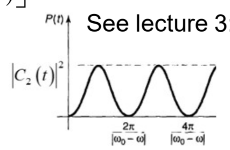
Derivation is similar to lecture 3 (slide 15), but $H_{21}$ is constant, independent of time:

$$C_2(t) = -\frac{i}{\hbar}\int_0^t \tilde{H}_{21} \exp(i\omega_0 t)\, dt = -\frac{i}{\hbar}\tilde{H}_{21} \frac{\exp(i\omega_0 t) - 1}{i\omega_0}$$

$$C_2(t) = -\frac{\tilde{H}_{21}}{\hbar}\frac{\exp(i\omega_0 t/2)}{\omega_0}\left[\exp(i\omega_0 t/2) - \exp(-i\omega_0 t/2)\right]$$

$$C_2(t) = -i\frac{2\tilde{H}_{21}}{\hbar}\frac{\sin(\omega_0 t/2)}{\omega_0}\exp(i\omega_0 t/2)$$

See lecture 3:




$$|C_2(t)|^2 = \frac{4|\tilde{H}_{21}|^2}{\hbar^2}\frac{\sin^2(\omega_0 t/2)}{\omega_0^2}$$

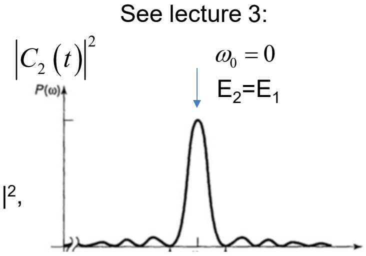
← Probability to find the system in state 2 after time t (energy transfer probability)



# Förster Resonance Energy Transfer

$$|C_2(t)|^2 = \frac{4|\tilde{H}_{21}|^2}{\hbar^2}\frac{\sin^2(\omega_0 t/2)}{\omega_0^2}$$

See lecture 3:



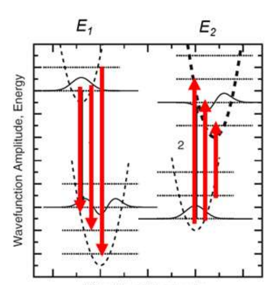

$$\omega_0 = \frac{E_2 - E_1}{\hbar}$$

Transfer rate maximizes when $E_2$=$E_1$

Also: However, $|C_2|^2$ oscillates as $|\sin[(E_2-E_1)t/h]|^2$,

→ Transitions occur only between states with the same energy **(Fermi golden rule)**. **(*resonance* energy transfer)**

Need to consider the contributions of all transitions with $E_1$ = $E_2$ →



# FRET: density of states

Transfer between two states : $\left|C_2^{(1)}(t)\right|^2 = \frac{4\left|\tilde{H}_{21}\right|^2}{\hbar^2}\frac{\sin^2(\omega_0 t/2)}{\omega_0^2}$

$\omega_0 \equiv \frac{E_2 - E_1}{\hbar} \equiv \frac{E_{21}}{\hbar} \qquad E_{21} \equiv E_2 - E_1$

Introduce:

*$\rho_{s1}(E)$ = density of initial states, $\rho_{s1}(E)dE$ is # of donor states between E…E+dE*
*$\rho_{s2}(E)$ = density of final states, $\rho_{s2}(E)dE$ is # of acceptor states between E…E+dE*

Probability to have pairs of states at $E_1$ and $E_2$: $\rho_{s1}(E_1)\rho_{s2}(E_2)\, dE_1 dE_2$

Account for all possible $E_1$,$E_2$ combinations:
Transfer probability in time *t* for all states of donor and acceptor

$$P = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |C_2|^2 \rho_{s2}(E_2)\rho_{s1}(E_1)dE_2 dE_1$$

Since $|C_2|^2$ is nonzero only when $E_2 \approx E_1 \rightarrow \rho_{s2}(E_2) \approx \rho_{s2}(E_1)$

$$P = \int_{-\infty}^{\infty}\left\{\int_{-\infty}^{\infty} |C_2|^2 dE_2\right\}\rho_{s2}(E_1)\rho_{s1}(E_1)dE_1$$

$$P = \int_{-\infty}^{\infty}\left\{\int_{-\infty}^{\infty} \frac{4\left|\tilde{H}_{21}\right|^2}{\hbar^2}\frac{\sin^2(E_{21}t/2\hbar)}{(E_{21}/\hbar)^2}dE_2\right\}\rho_{s2}(E_1)\rho_{s1}(E_1)dE_1$$

Integral over E:
Replace $\omega_0 \equiv \frac{E_{21}}{\hbar}$

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# Rate of FRET

Evaluate integral: $P = \int_{-\infty}^{\infty}\left\{\int_{-\infty}^{\infty} \frac{4\left|\tilde{H}_{21}\right|^2}{\hbar^2}\frac{\sin^2(E_{21}t/2\hbar)}{(E_{21}/\hbar)^2}dE_2\right\}\rho_{s2}(E_1)\rho_{s1}(E_1)dE_1$

$$P \approx \int_{-\infty}^{\infty} \frac{4\left|\tilde{H}_{21}\right|^2}{\hbar^2}\underbrace{\boxed{\left\{\int_{-\infty}^{\infty} \frac{\sin^2(E_{21}t/2\hbar)}{(E_{21}/\hbar)^2}dE_2\right\}}}_{=\frac{\pi\hbar t}{2}}\rho_{s2}(E_1)\rho_{s1}(E_1)dE_1$$

$E_{21} = E_2 - E_1$
$dE_{21} = dE_2$

$H_{21}$ for transition at energy $E_1$

Table integral
$\int_{-\infty}^{\infty}\frac{\sin^2(ax)}{x^2}dx = \pi|a|$

$$P \approx \int_{-\infty}^{\infty} \frac{4\left|\tilde{H}_{21}\right|^2}{\hbar^2}\frac{\pi\hbar t}{2}\rho_{s2}(E_1)\rho_{s1}(E_1)dE_1 = \frac{2\pi t}{\hbar}\int_0^{\infty}\left|\tilde{H}_{21}\right|^2 \rho_{s2}(E_1)\rho_{s2}(E_1)dE_1$$

Energy transfer rate (in transfer events per second):

$$k_{rt} = \frac{P}{t} \approx \frac{2\pi}{\hbar}\int_0^{\infty}\left|\tilde{H}_{21}\right|^2 \rho_{s2}(E_1)\rho_{s2}(E_1)dE_1$$

(can replace $E_1$ with E)

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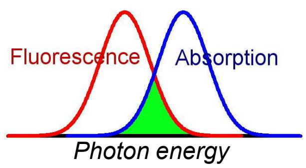
# FRET: spectral overlap

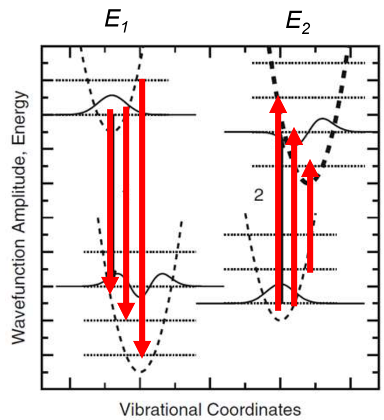
$$\boxed{k_{rt} \approx \frac{2\pi}{\hbar}\int_{-\infty}^{\infty}\left|\tilde{H}_{21}\right|^2 \rho_{s1}(E)\rho_{s2}(E)\,dE}$$

For constant $H_{21}$ can rewrite:

$$k_{rt} \approx \frac{2\pi}{\hbar}\left|\tilde{H}_{21}\right|^2 \boxed{\int_{-\infty}^{\infty}\rho_{s1}(E)\rho_{s2}(E)\,dE}$$

Spectral overlap integral




Absorption spectrum: probes density of states of the *acceptor*
Fluorescence: probes density of states of the *donor*

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